

Abstract

Title: Beyond Acute Biomarkers: Predicting Two-Month Functional Disability in ICU Oncologic Patients Using Frailty Embeddings of Chronic and Sociodemographic Features

Authors: Claudia Bollettini, Carlo Rizzo, Elisa Totis

Purpose

Predicting functional decline in seriously ill hospitalized cancer patients remains a major clinical challenge¹. Using the SUPPORT2 dataset², we predicted 2-month functional disability (independent vs impaired, excluding deceased patients) and evaluated whether latent feature compression of chronic and sociodemographic variables preserves predictive performance.

Materials and methods

We analyzed 1,371 SUPPORT2 cancer patients (train: n=1,165; test: n=206) to predict binary functional disability. To prevent leakage, imputation (physiological values for clinical variables³, Gower distance for income), scaling, one-hot encoding, and hybrid resampling (random under-sampling plus SMOTE) were embedded in a 5x5 nested cross-validation. We trained Logistic Regression, Decision Tree, Random Forest, and XGBoost, tuning hyperparameters for AUC via Grid Search. Four feature strategies were compared: (1) baseline features; (2) Recursive Feature Elimination (RFE); dimensionality reduction of chronic/sociodemographic variables via (3) PCA or (4) Autoencoders, concatenated with acute features. Test set performance was evaluated using AUC, accuracy, precision, and F1-score, while SHAP values assessed model interpretability.

Results

Test-set (n=206) discrimination was comparable across pipelines (Baseline AUC=0.624, RFE=0.603, PCA=0.608, Autoencoder=0.615). Across all approaches, Logistic Regression showed the best balance between performance, overfitting control, and stability. Among reduced-feature methods, the Autoencoder pipeline combining latent chronic/sociodemographic embeddings with acute clinical variables achieved the strongest overall trade-off, approaching baseline performance while substantially compressing the feature space.

Conclusions

Our findings suggest that chronic and sociodemographic variables can be substantially compressed into lower-dimensional representations without meaningful loss of predictive information, indicating that much of the relevant signal may lie in a compact latent structure rather than the full original feature space. These results support the feasibility of feature compression strategies for simplifying multimodal clinical datasets while maintaining comparable predictive performance across pipelines. However, the study is limited by the use of a binary and partially operator-dependent outcome, feature loss related to missingness, and the trade-off between reduced feature complexity and decreased interpretability introduced by latent representation methods.

		CV (train)	CV (validation)	Held out (test)
Baseline	AUC ($\pm$ std)	0.668 $\pm$ 0.008	0.639 $\pm$ 0.030	0.6244
	Accuracy ($\pm$ std)	0.616 $\pm$ 0.021	0.606 $\pm$ 0.028	0.6165
RFE Pipeline	AUC ($\pm$ std)	0.657 $\pm$ 0.007	0.645 $\pm$ 0.030	0.6039
	Accuracy ($\pm$ std)	0.627 $\pm$ 0.046	0.591 $\pm$ 0.045	0.5534
PCA Pipeline	AUC ($\pm$ std)	0.657 $\pm$ 0.009	0.626 $\pm$ 0.025	0.6084
	Accuracy ($\pm$ std)	0.603 $\pm$ 0.024	0.583 $\pm$ 0.041	0.6117
AE Pipeline	AUC ($\pm$ std)	0.660 $\pm$ 0.008	0.634 $\pm$ 0.027	0.6156
	Accuracy ($\pm$ std)	0.602 $\pm$ 0.025	0.585 $\pm$ 0.036	0.6214

Table: Results of the best model of all pipelines (Logistic Regression) in terms of AUC and Accuracy for Train and Test splits of the CV and Held out Test Set.

References

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